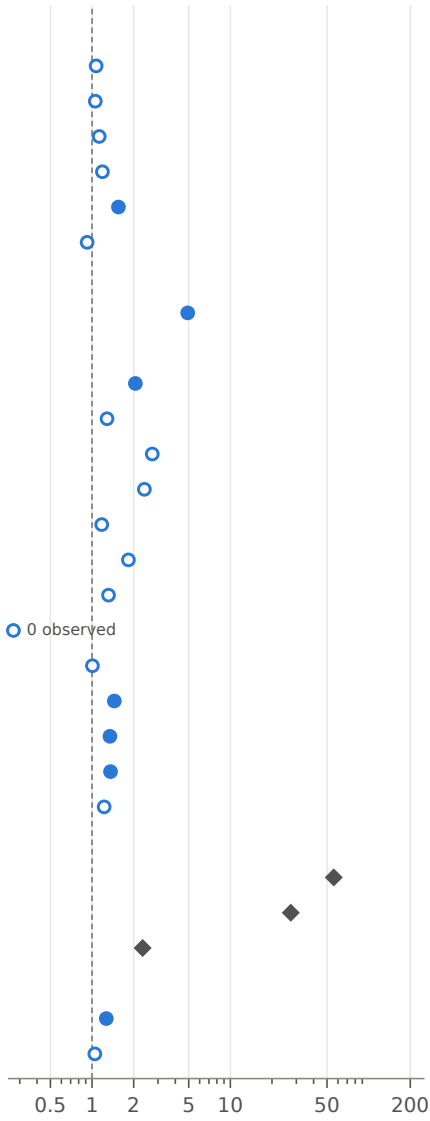
**A**

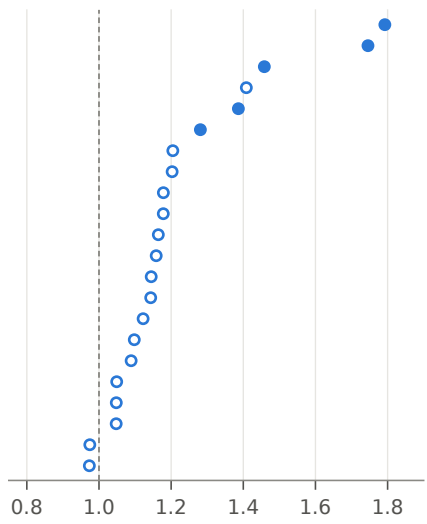



| | obs / exp loci | p, matched null loci | MAGMA p (gene level) |
|---|---|---|---|
| **Mendelian disease genes** | | | |
| Any Mendelian (OMIM or Orphanet) | 123 / 114.9 | 0.16 | 0.75 |
| OMIM | 118 / 111.8 | 0.22 | 0.74 |
| Orphanet (assessed) | 111 / 98.5 | 0.054 | 0.92 |
| ClinGen moderate–definitive | 77 / 64.8 | 0.041 | 0.39 |
| Autosomal dominant | 80 / 51.6 | $1.1\times10^{-5}$ | 0.28 |
| Autosomal recessive | 74 / 80.3 | 0.84 | 0.99 |
| **Positive control** | | | |
| GO: circadian rhythm | 22 / 4.5 | $8.4\times10^{-10}$ | 0.0019 |
| **Pre-specified phenotypes** | | | |
| Sleep disturbance | 34 / 16.5 | $3.5\times10^{-5}$ | 0.031 |
| without neurodevelopmental features | 4 / 3.1 | 0.38 | 0.19 |
| Sleep–wake cycle / melatonin | 3 / 1.1 | 0.098 | 0.23 |
| Prolonged QT interval | 3 / 1.3 | 0.13 | 0.86 |
| Arrhythmia | 17 / 14.5 | 0.28 | 0.84 |
| Hypertrophic cardiomyopathy | 13 / 7.1 | 0.024 | 0.88 |
| Cardiomyopathy | 21 / 16.0 | 0.11 | 0.58 |
| Reduced circulating complement | 0 / 0.90 | 1.00 | 0.60 |
| Immunodeficiency | 5 / 5.0 | 0.56 | 0.99 |
| Global developmental delay | 76 / 52.5 | $1.7\times10^{-4}$ | 0.17 |
| Intellectual disability | 71 / 52.7 | 0.0029 | 0.089 |
| Seizure | 70 / 51.5 | 0.0022 | 0.57 |
| Short stature | 44 / 36.0 | 0.082 | 0.22 |
| **Post hoc** | | | |
| Monoamine neurotransmitter disorders | 4 / 0.07 | $6.7\times10^{-7}$ | 0.032 |
| Decreased CSF homovanillic acid | 3 / 0.11 | $1.5\times10^{-4}$ | 0.092 |
| Complement deficiency genes (IUIS) | 2 / 0.86 | 0.21 | 0.52 |
| **Rhythmic expression** | | | |
| Rhythmic in ≥3 mouse tissues | 105 / 82.8 | 0.0014 | 0.0086 |
| Human day–night (GTEx) | 44 / 42.0 | 0.38 | 0.011 |

Null loci matched on MAF, LD, gene count and distance

● FDR q < 0.05 within set family ○ FDR q ≥ 0.05 ◆ post hoc set (q not shown)

**B**




| | obs / exp loci | p, matched null loci | MAGMA p (gene level) |
|---|---|---|---|
| Voice | 20 / 11.2 | 0.0078 | 0.47 |
| Breast | 21 / 12.0 | 0.0092 | 0.68 |
| Endocrine system | 50 / 34.3 | 0.0029 | 0.79 |
| Prenatal development or birth | 36 / 25.6 | 0.020 | 0.60 |
| Limbs | 82 / 59.1 | $4.4\times10^{-4}$ | 0.72 |
| Genitourinary system | 80 / 62.4 | 0.0050 | 0.79 |
| Head or neck | 94 / 78.0 | 0.015 | 0.72 |
| Digestive system | 82 / 68.2 | 0.025 | 0.63 |
| Neoplasm | 27 / 22.9 | 0.21 | 0.99 |
| Musculoskeletal system | 102 / 86.6 | 0.020 | 0.73 |
| Ear | 70 / 60.1 | 0.076 | 0.77 |
| Growth abnormality | 72 / 62.1 | 0.077 | 0.94 |
| Cardiovascular system | 76 / 66.4 | 0.087 | 0.95 |
| Nervous system | 106 / 92.7 | 0.042 | 0.57 |
| Integument | 68 / 60.6 | 0.14 | 0.95 |
| Metabolism/homeostasis | 68 / 61.9 | 0.19 | 0.92 |
| Eye | 83 / 76.2 | 0.18 | 0.58 |
| Immune system | 49 / 46.7 | 0.37 | 0.78 |
| Constitutional symptom | 32 / 30.5 | 0.42 | 0.99 |
| Respiratory system | 54 / 51.5 | 0.37 | 0.72 |
| Abnormal cellular phenotype | 18 / 18.5 | 0.59 | 0.95 |
| Blood and blood-forming tissues | 36 / 37.0 | 0.60 | 0.94 |